Assignment 02
Bayesian Models & Data Analysis

Hemakshi Bansal

February 5, 2026

Part 1: Continuous Random Variables

1.1 The Shooter Experiment

The shooter tries to hit a target 5 times. The margins of error (in cm) are: $5.76, 4.56, 2.33, 6.32, 5.4$.

(a) Describe the random variable X The random variable X maps the sample space to a discrete space Ω_x. A trial is a "success" if the margin is ≤ 5 cm, and a "failure" if the margin is > 5 cm. X represents the total number of successes in 5 trials.

Analyzing the observed data:

- 5.76 (> 5) $\rightarrow$ Failure
- 4.56 (≤ 5) $\rightarrow$ Success
- 2.33 (≤ 5) $\rightarrow$ Success
- 6.32 (> 5) $\rightarrow$ Failure
- 5.40 (> 5) $\rightarrow$ Failure

Total successes observed $= 2$.
However, as a function $X(w)$ generally: X is the count of margins $m \in w$ such that $m \leq 5$. The possible values for X are $\{0, 1, 2, 3, 4, 5\}$.

(b) Define a probability distribution for X The random variable X follows a **Binomial Distribution** with parameters $n = 5$ and probability of success p. The probability mass function (PMF) is defined as:

$$f(x) = P(X = x) = \binom{5}{x} p^x (1-p)^{5-x}$$

where:

- $x \in \{0, 1, 2, 3, 4, 5\}$ is the number of successes.
- p is the probability that the shooter misses by ≤ 5 cm in a single trial.

1.2 Normal Distribution Probabilities

Given PDF: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$.

(a) Probability density of obtaining $x = 0$ given $\mu = 1, \sigma = 1$

$$f(0) = \frac{1}{1\sqrt{2\pi}} e^{-\frac{(0-1)^2}{2(1)^2}} = \frac{1}{\sqrt{2\pi}} e^{-0.5} \approx \mathbf{0.242}$$

(b) Probability density of obtaining $x = 1$ given $\mu = 0, \sigma = 1$

$$f(1) = \frac{1}{1\sqrt{2\pi}} e^{-\frac{(1-0)^2}{2(1)^2}} = \frac{1}{\sqrt{2\pi}} e^{-0.5} \approx \mathbf{0.242}$$

(Note: The density is identical because the distance from the mean is 1 unit in both cases).

(c) Finding Interval Probability

Given:

1. $P(x_1 \leq X \leq x_2) = 0.3$
2. $P(x_1 \leq X \leq x_3) = 0.45$

Assuming $x_1 < x_2 < x_3$, the interval $[x_1, x_3]$ is the union of $[x_1, x_2]$ and $[x_2, x_3]$.

$$P(x_2 \leq X \leq x_3) = P(x_1 \leq X \leq x_3) - P(x_1 \leq X \leq x_2)$$

$$P(x_2 \leq X \leq x_3) = 0.45 - 0.3 = \mathbf{0.15}$$

1.3 Beta Distribution Mode

PDF: $f(x) \propto x^{\alpha-1}(1-x)^{\beta-1}$ for $x \in [0, 1]$. To find the mode, we maximize $f(x)$ (or equivalently $\ln f(x)$) by setting the derivative to zero.

$$\ln f(x) = C + (\alpha - 1)\ln x + (\beta - 1)\ln(1 - x)$$

$$\frac{d}{dx} \ln f(x) = \frac{\alpha - 1}{x} - \frac{\beta - 1}{1 - x} = 0$$

$$(\alpha - 1)(1 - x) = (\beta - 1)x$$

$$\alpha - 1 - \alpha x + x = \beta x - x$$

$$\alpha - 1 = x(\alpha + \beta - 2)$$

$$\text{Mode } x = \frac{\alpha - 1}{\alpha + \beta - 2}$$

1.4 Standard Normal Distribution

PDF: $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

(a) Mean and Variance By comparing to the general normal form, this is the Standard Normal Distribution.

- Mean (μ) = $\mathbf{0}$
- Variance (σ^2) = $\mathbf{1}$

(b) Probability density of obtaining $X = 0$

$$f(0) = \frac{1}{\sqrt{2\pi}}e^0 = \frac{1}{\sqrt{2\pi}} \approx \mathbf{0.3989}$$

(c) Probability of score less than 0 ($X < 0$) Since the distribution is symmetric around the mean $\mu = 0$:

$$P(X < 0) = \mathbf{0.5}$$

(d) Probability of obtaining $X = 0$ For any continuous random variable, the probability of obtaining an exact point value is zero.

$$P(X = 0) = \mathbf{0}$$

Part 2: Distribution of Distances in Natural Languages

Poisson-like PMF: $f(k, \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$

2.1 Highest frequency of distance 0

We calculate $f(0)$ for different λ.

$$f(0) = \frac{\lambda^0 e^{-\lambda}}{0!} = e^{-\lambda}$$

Since $e^{-\lambda}$ is a decreasing function (smaller λ yields larger probability), and we are given $\lambda_3 < \lambda_1 < \lambda_2$:
Czech (λ_3) has the highest number of word pairs with linear distance 0.

2.2 Calculating specific word pair counts ($k = 1$)

Formula for probability: $P(k = 1) = \frac{\lambda^1 e^{-\lambda}}{1!} = \lambda e^{-\lambda}$. Multiply probability by Total Words (N) to get the count.

English ($\lambda_1 = 0.7, N = 10^5$):

$$P(1) = 0.7e^{-0.7} \approx 0.7 \times 0.4966 = 0.3476$$

$$\text{Count} = 0.3476 \times 10^5 = \mathbf{34{,}761}$$

German ($\lambda_2 = 0.8, N = 10^7$):

$$P(1) = 0.8e^{-0.8} \approx 0.8 \times 0.4493 = 0.3595$$

$$\text{Count} = 0.3595 \times 10^7 = \mathbf{3{,}594{,}603}$$

Czech ($\lambda_3 = 0.6, N = 10^8$):

$$P(1) = 0.6e^{-0.6} \approx 0.6 \times 0.5488 = 0.3293$$

$$\text{Count} = 0.3293 \times 10^8 = \mathbf{32{,}928{,}700}$$

2.3 Random Structure Comparison

The random structure has $\lambda = 2$. Natural languages (as seen in 2.2) have $\lambda < 1$ (0.6, 0.7, 0.8).
Difference: The random structure has a much higher average distance ($\lambda = 2$) between connected nodes. Natural languages are optimized for "dependency minimization," meaning related words tend to be much closer together (often adjacent, distance 0 or 1) compared to a random graph.

Part 3: The Likelihood Function

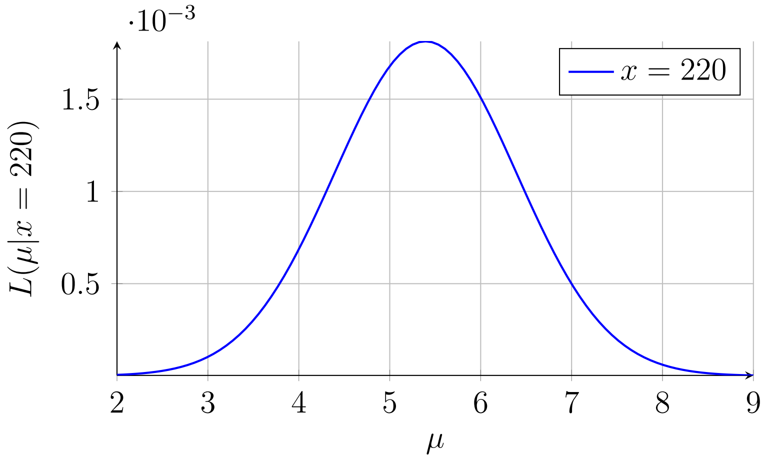
Model: $f(x, \mu) = \frac{1}{x\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2}}$ (Log-Normal, assuming $\sigma = 1$ based on formula provided). Observed Data: $303.25, 443, 220, 560, 880$.

3.1(a) Likelihood Graph for fixed $x = 220$

The likelihood function is given by:

$$L(\mu|x = 220) = \frac{1}{220\sqrt{2\pi}} e^{-\frac{(\ln 220 - \mu)^2}{2}}$$

Below is the plot of this function with respect to μ.

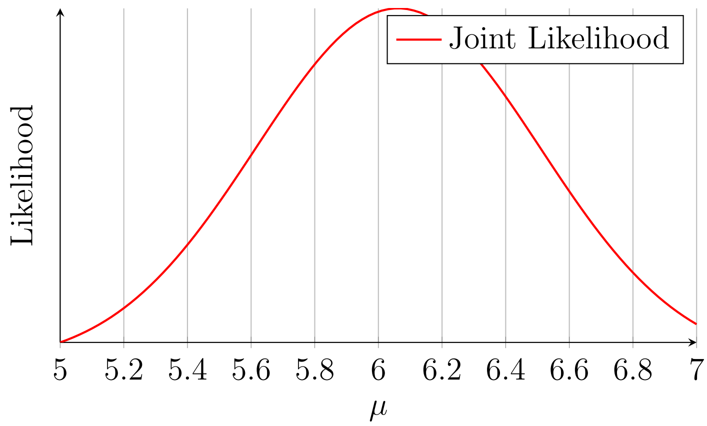


3.1(b) Likelihood Graph for observed sample

The likelihood function for the sample $x = [303.25, 443, 220, 560, 880]$ is the product of individual densities:

$$L(\mu|x_{obs}) = \prod_{i=1}^{5} \frac{1}{x_i\sqrt{2\pi}} e^{-\frac{(\ln x_i - \mu)^2}{2}}$$

This results in a much narrower curve centered around the mean of the logs.



3.1(c) Maximum Likelihood Estimation

To maximize the likelihood, we maximize the log-likelihood, which is equivalent to minimizing the sum of squared errors: $\sum(\ln x_i - \mu)^2$. The maximum occurs at the mean of the log-values: $\hat{\mu} = \frac{1}{n}\sum \ln x_i$.

Data Logs ($\ln x$):

- $\ln(303.25) \approx 5.715$
- $\ln(443) \approx 6.094$
- $\ln(220) \approx 5.394$
- $\ln(560) \approx 6.328$
- $\ln(880) \approx 6.780$

$$\text{Sum} \approx 30.311$$
$$\hat{\mu} \approx \frac{30.311}{5} \approx \mathbf{6.06}$$

Part 4: Distributions in R

Question 4.1

We are given a normal distribution with mean $\mu = 500$ and standard deviation $\sigma = 100$. The function `pnorm(x,mean,sd)` computes $P(X \leq x)$ for a normally distributed random variable X.

(a) We want the probability of obtaining a value ≤ 700:

$$P(X \leq 700) = \text{pnorm}(700, 500, 100)$$

Standardizing using $Z = \frac{X-\mu}{\sigma}$:

$$z = \frac{700 - 500}{100} = 2$$

So,

$$P(X \leq 700) = P(Z \leq 2) \approx 0.9772$$

(b) We want the probability of obtaining a value ≥ 900:

$$P(X \geq 900) = 1 - P(X \leq 900)$$

Standardizing:

$$z = \frac{900 - 500}{100} = 4$$

$$P(X \geq 900) = 1 - P(Z \leq 4) \approx 1 - 0.99997 \approx 0.00003$$

This is extremely small, which makes sense given how far 900 is from the mean.

(c) We want the probability that the value lies between 200 and 800:

$$P(200 \leq X \leq 800) = P(X \leq 800) - P(X \leq 200)$$

Standardizing:

$$z_1 = \frac{800 - 500}{100} = 3, \quad z_2 = \frac{200 - 500}{100} = -3$$

$$P(200 \leq X \leq 800) = P(Z \leq 3) - P(Z \leq -3)$$

$$\approx 0.99865 - 0.00135 = 0.9973$$

Question 4.2

We are given a normal distribution with:

$$\mu = 650, \quad \sigma = 125$$

The probability density at $x = 550$ is:

$$f(550) = \frac{1}{125\sqrt{2\pi}} e^{-\frac{(550-650)^2}{2(125)^2}}$$

Simplifying the exponent:

$$\frac{550 - 650}{125} = -0.8$$

$$f(550) = \frac{1}{125\sqrt{2\pi}} e^{-0.32}$$

This value represents a density, not a probability.

$$f(550) \approx 0.0023$$

Question 4.3

We generate $n = 10000$ samples using:

$$x_{\text{sim}} \sim \mathcal{N}(300, 200^2)$$

By the Law of Large Numbers, as n becomes large:

- The sample mean approaches 300
- The sample standard deviation approaches 200
- The minimum and maximum vary randomly but spread roughly over a few standard deviations

So computed values from x_{sim} should be close to:

$$\text{mean} \approx 300, \quad \text{sd} \approx 200$$

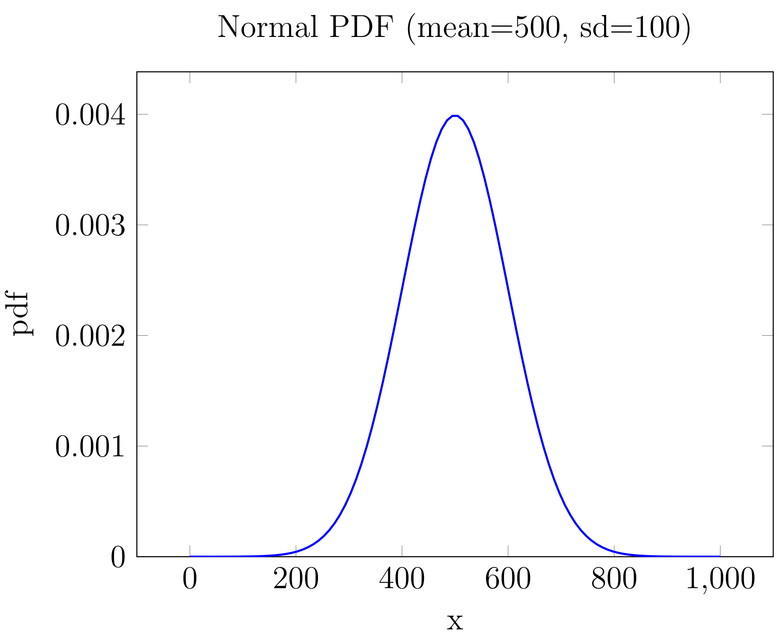


Figure 1: Figure for Q 4